Durganshu Mishra

e-mail	durganshu.mishra@tum.de
Mobile	+4915730109560
Github	github.com/Durganshu
LinkedIn	linkedin.com/in/durganshu/
Address	Enzianstraße 7 | 85748 Garching bei München

Relevant Experience

July 2023 - today

Chair of Computer Architecture and Parallel Systems, TU Munich

Hilfswissenschaftler (Research Assistant)

- Involved in performance optimization and identifying potential memory leaks within the sys-sage codebase
- Reduced valgrind-reported memory leaks by ~99% in sys-sage (See github.com/Durganshu/sys-sage)
- Improving the portability of mitos for Intel-based computing systems (See github.com/Durganshu/mitos)

May 2022 - May 2023

Intel Deutschland GmbH, Munich (Germany)

Working Student - Cloud Graphics and Gaming

- Performance analysis and benchmarking of Vulkan and OpenGL-based graphics applications on Intel GPUs and CPUs
- Extended the features of graphics applications for cross-OS (Linux and Windows) and cross-API compatibilities

August 2020 - August 2021

Zeus Numerix Pvt. Ltd., Pune (India)

CAE engineer (Development)

- Involved in rigorous problem-solving and software development (using C++, Python, and MATLAB) in the domains of computational electromagnetism, radar signature prediction, and target detection
- Implemented specific features to the in-house developed CAE products and customized them as per clientele's needs

Academic Career

October 2021 - today

Technische Universität München, Germany

Master of Science in Computational Science and Engineering

Coursework: Advanced Programming, Scientific Computing, Computer Architectures, Parallel Programming, Numerical Algo. for HPC, CFD.

Jan. 2023 – June 2023

École polytechnique, Paris

Visiting student (Virtual) - EuroTeQ

Coursework: Distributed Applications

August 2016 - July 2020 **PDPM ndian Institute of Information Technology, Design and Manufacturing, Jabalpur, India**
Bachelor of Technology in Mechanical Engineering
Bachelor's Thesis: Numerical Investigation of Jet Impingement Quenching

Relevant Projects

June 2023 - July 2023 **Abalone game**

- A C++ based implementation of an Abalone board game
- Implemented Minimax and parallelized Alpha-beta pruning as search strategies (See github.com/Durganshu/Abalone-Game)
- **Technologies:** C++, Python, OpenMP (tasking), Git, Linux

April 2022 - July 2022 **Fluidchen- EM**

- A C++ based Computational Fluid Dynamics solver
- Integrated with MPI for parallelization on distributed systems
- Integrated with preCICE for coupling with other solvers (See github.com/Durganshu/Fluidchen-EM)
- **Technologies:** C++, MPI, preCICE, Git, Linux

May 2022 **Parallelized Jacobi solver**

- Implementation of the Jacobi method on a discretized 5-point stencil.
- Supports shared, distributed and hybrid memory parallelism (See github.com/Durganshu/Parallelized-Jacobi-Method)
- **Technologies:** C++, OpenMP, MPI, Git, Linux

April 2022 **DeepBio**

- An AI-based tool to help seniors with a voice-based system
- Useful for automated biography writing and questionnaires (See https://github.com/Durganshu/DeepBio)
- **Technologies:** Python, PyQt, OpenAI, Codex

Knowledge and skills

Programming	C, C++, Python, Julia, MATLAB
Frameworks and API	OpenMP, MPI, Pandas, NumPy, SciPy, Matplotlib, Pytorch
Documentation	MS Office, LaTex
Languages	English (Fluent), German (Basic knowledge - A2), Hindi (Native)